

Prognostic value of three-tiered scoring system for lymph-vascular space invasion in endometrial cancer: A systematic review and meta-analysis

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HIGHLIGHTS

- Survival was worse in EC cases with substantial or focal LVSI compared to those without LVSI.
- The substantial LVSI group had a worse prognosis than the focal LVSI group.
- A three-tiered LVSI scoring system may predict EC prognosis.

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ABSTRACT

Objective. To investigate the impact of lymph-vascular space invasion (LVSI) status on the prognosis of endometrial cancer (EC) according to a three-tiered scoring system for LVSI.

Methods. PubMed, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), International Clinical Trials Registry Platform (ICTRP), and Clinical [Trials.gov](http://trials.gov) were searched from inception to September 1st, 2023. The analysis was conducted using STATA 16.0.

Results. A total of 9 studies with 4456 EC patients were included in the analysis. No LVSI was found in 72% of EC patients (95% CI 0.65–0.79), while focal and substantial LVSI were present in 16% (95% CI 0.11–0.21) and 13% (95% CI 0.08–0.18) of patients, respectively. Compared to the no LVSI group, the focal and substantial LVSI groups had poorer overall survival (for focal LVSI: HR 1.33, 95% CI 1.02–1.74; for substantial LVSI: HR 2.51, 95% CI 1.61–3.90), poorer disease-free survival (for substantial LVSI: HR 2.86, 95% CI 1.21–6.77), and an increased risk of recurrence, including pelvic recurrence (for focal LVSI: HR 2.05, 95% CI 1.03–4.07; for substantial LVSI: HR 6.06, 95% CI 3.31–11.08), distant recurrence (for focal LVSI: HR 2.04, 95% CI 1.42–2.92; for substantial LVSI: HR 3.36, 95% CI 2.35–4.793), and lymph node involvement (for focal LVSI: OR 3.52, 95% CI 1.339–9.34; for substantial LVSI: OR 5.42, 95% CI 2.78–10.58). Substantial LVSI was more prone to pelvic recurrence (HR 1.82, 95% CI 1.05–3.15) and distant recurrence (HR 2.21, 95% CI 1.48–3.28) than focal LVSI.

Conclusions. EC patients with focal and substantial LVSI had poorer survival, recurrence, and a higher incidence of lymph node metastasis than patients without LVSI. The substantial LVSI group was associated with even worse prognosis than the focal LVSI group. The three-tiered LVSI scoring system might effectively predict the prognosis of EC and guide clinical decision-making.

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1. Introduction

Endometrial cancer (EC) is the most common gynecological malignancy in the United States with 65,950 new cases projected for 2022 and 12,550 related deaths [1,2]. Studies indicate that 68%, 21%, and 8% of EC patients have lesions limited to the uterus, regional disease, and distant disease, respectively [3]. The standard treatment for EC is surgery, which involves a total hysterectomy and bilateral salpingo-oophorectomy. Clinicopathological risk factors are identified for EC including advanced age, tumor grade, size, histological type, depth of uterine myometrial invasion, presence of cervical interstitial infiltration, and lymphatic vessel interstitial infiltration (LVSI). These risk factors can be grouped into low, low-intermediate risk, high-intermediate risk, or high risk for recurrence [4,5].

According to reports, LVSI is present in approximately 8–10% of FIGO stage I EC patients [6] and has been with a higher risk of lymph node metastasis, recurrence, and poor overall survival [7,8]. Despite this, guidelines do not consistently consider LVSI as a prognostic factor to inform postoperative treatment for EC. The European Society for Medical Oncology (ESMO), European Society for Radiotherapy & Oncology (ESTRO), and European Society of Gynecological Oncology (ESGO) use LVSI as a risk factor for tumor staging [5]. On other hand, the National Comprehensive Cancer Network (NCCN) 2023 still provides treatment recommendations based on the International Federation of Gynecology and Obstetrics (FIGO) 2009 staging, lacking LVSI as a component for the staging standard [1].

At first, LVSI was categorized as present or absent. However, Hachisuga et al. proposed a three-tiered scoring system in 1999 (no, focal, substantial) and found that this semi-quantitative grading was associated with prognosis and lymph node metastasis [9]. Bosse et al. found, in their research (PORTEC-1 and -2), that a four-tiered system (no, minimal, moderate, and prominent) did not provide additional value compared to three-tiered system. The three-tiered system was found to be better at predicting prognosis than the two-tiered system [6]. Recently, FIGO published a new staging system that includes the three-tiered scoring system for LVSI to better inform treatment decisions [10]. This study aimed to investigate the impact of different LVSI

status on the prognosis of EC based on the three-tiered scoring system for LVSI.

2. Methods

2.1. Protocol registration

This meta-analysis was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines and was registered with the International Prospective Register of Systematic Reviews (CRD42023451793) [11].

2.2. Eligibility criteria

All potentially eligible studies including cross-sectional studies, longitudinal cohort studies, case reports and series case reports published in English were considered. The inclusion criteria were as follows: (a) investigate the prognostic role of a three-tiered scoring system of LVSI in early-stage EC with low-risk pathologic features, (b) provide effect data including odds ratios (OR)/relative risk (RR)/hazard (HR) with 95% confidence intervals (CI) or enable calculation of these data, (c) if data subsets had been published in more than one article, only the largest sample size one was included. The exclusion criteria were as follows: (a) redundant publications, (b) incomplete data, (c) conference abstracts and reviews, (d) if data subsets had been published in more than one article, only the largest sample size one was included.

2.3. Search strategy and study selection

PubMed, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), International Clinical Trials Registry Platform (ICTRP), and Clinical Trials.gov were searched from inception to September 1st, 2023. The reference lists of the published reviews and retrieved articles were checked for additional trials. Predefined search strings were as follows: “lymphovascular space invasion”, “lymph-vascular space invasion”, “LVSI”, “survival”, “outcome”, “prognosis”, “recurrence”, “relapse” and “endometrial cancer”.

Two researchers (HC and LH) independently screened the titles and abstracts to assess the eligibility of the studies. After initial selection, the full texts of all potential articles were independently read by two researchers (HC and LH) for further evaluation. Disagreements between authors were resolved by discussion with the XT.

2.4. Data extraction

Data were extracted by two independent reviewers (HC and YC) in duplicates and recorded in a standardized database. We used a predefined extraction form including LVSI rates, countries, study quality and outcomes. Review authors were blinded to the trial authors, institutions, sources of funding, and acknowledgments. A double data entry was conducted. We tried to acquire missing data by contacting the author via email, but no reply was received.

2.5. Risk of bias assessment

Two reviewers (LH and YC) independently assessed the quality of the included studies. We resolved differences by discussion and, if no consensus was reached, by involving a third review author (AZ). Cohort studies included in the prognosis analysis were assessed using the Newcastle–Ottawa Scale (NOS) based on three categories: selected cases, comparability of groups, and assessment of outcomes. Studies that were awarded six or more stars were classified as having high quality.

2.6. Statistical analysis

STATA 16.0 (Statacorp, USA) was used for meta-analysis. Hazard ratios (HRs) with 95% confidence intervals (CIs) was used to combine data regarding survival and recurrence outcomes. For studies reporting survival data only in the form of Kaplan–Meier curves, Engauge Digitizer 4.1 was used to extract survival data, and HRs and CIs were calculated according to the methods reported in literature. The LVSI rate (%) was investigated by logistic regression analysis. P values of <0.05 was considered statistically significant for the meta-analysis. The heterogeneity between studies was assessed using I^2 test: $I^2 < 30\%$ was considered low heterogeneity, $30\% < I^2 < 50\%$ was considered moderate heterogeneity, and $I^2 \geq 50\%$ was considered high heterogeneity. When there was substantial heterogeneity, we used random-effects model to combine data. Instead, fix-effects model was used. Because fewer than 10 articles were included, we did not conduct publication bias analysis. For the results that could not be meta-analyzed, we reported them in a narrative manner.

3. Results

3.1. Study selection and characteristics

The study selection process was briefly depicted in Fig. 1. After removing duplicates, 916 articles were retrieved and screened based on their titles and abstracts. Later, 12 full texts were procured for further assessment, of which 3 articles were excluded after reading the full texts. Ultimately, the analysis included 9 studies with 4456 participants [6,12–19]. The general characteristics of these studies were presented in Table 1. Eight were retrospective studies and one was pooled analysis study. All included studies received a rating of six or more stars according to the NOS criteria.

3.2. Rates of LVSI

Based on the three-tiered scoring system, the results of the study showed that 72% of EC patients had no LVSI (95% CI 0.65–0.79, $p = 0.000$, $I^2 = 96.5\%$), and the incidence of focal and substantial LVSI were 16% (95% CI 0.11–0.21, $p = 0.000$, $I^2 = 95.1\%$) and 13% (95% CI 0.08–0.18, $p = 0.000$, $I^2 = 96.8\%$), respectively (Fig. 2) [6,12–19].

3.3. Survival outcomes according to the three-tiered scoring system for LVSI

The meta-analysis results revealed that, compared to the no LVSI group, both focal and substantial LVSI groups had poorer overall survival (OS), as evidenced by a HR of 1.33 (95% CI 1.02–1.74, $p = 0.038$, $I^2 = 0.0\%$) for focal LVSI, and HR of 2.51 (95% CI 1.61–3.90, $p = 0.000$, $I^2 = 50.0\%$) for substantial LVSI [6,12,14,17,18]. Disease-free survival (DFS) in the substantial LVSI group worsened (HR 2.86, 95% CI 1.21–6.77, $p = 0.017$, $I^2 = 77.3\%$), while there seemed to be no difference in the focal LVSI group (HR 1.22, 95% CI 0.66–2.26, $p = 0.524$, $I^2 = 56.7\%$) (Fig. 3) [12,16–18].

3.4. Recurrence outcomes according to the three-tiered scoring system for LVSI

The meta-analysis results showed that, compared to the no LVSI group, focal and substantial LVSI were associated with higher rates of local (for substantial LVSI: HR 2.23, 95% CI 1.04–4.77, $p = 0.039$, $I^2 = 0.0\%$), pelvic (for focal LVSI: HR 2.05, 95% CI 1.03–4.07, $p = 0.040$, $I^2 = 0.0\%$; for substantial LVSI: HR 6.06, 95% CI 3.31–11.08, $p = 0.000$, $I^2 = 0.0\%$), and distant recurrence (for focal LVSI: HR 2.04, 95% CI 1.42–2.92, $p = 0.000$, $I^2 = 0.0\%$; for substantial LVSI: HR 3.36, 95% CI

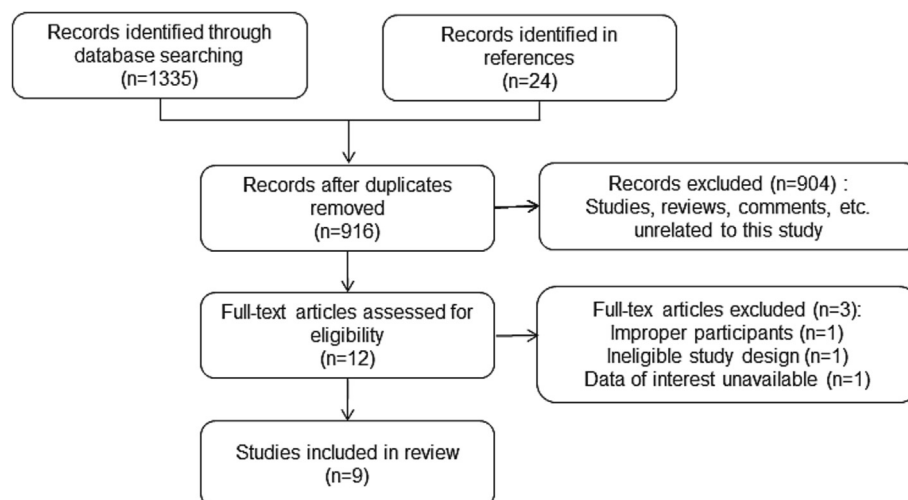


Fig. 1. Flowchart of the study selection.

Table 1
The basic characteristics of the included studies.

Study	Country	Study type	FIGO stage	Number of participants	Number of participants with no LVSI	Number of participants with focal LVSI	Number of participants with substantial LVSI
Barnes 2021 [1,2]	Canada	Retrospective study	I–II	291	213	45	33
Bosse 2015 [6]	Netherlands	Pooled analysis study	I	926	797	85	44
Matsuo 2015 [13]	California	Retrospective study	I–III	300	245	44	11
Peters 2022 [14]	Denmark	Retrospective study	I–II	305	261	25	19
			III–IV		35	9	27
Pifer 2020 [15]	America	Retrospective study	I	438	295	73	70
Pifer 2023 [16]	America	Retrospective study	I	335	211	65	59
Tortorella 2021 [17]	America	Retrospective study	I	524	445	57	22
Restaino 2020 [18]	Italy	Retrospective multicenter study	I–III	1258	846	160	252
Jaishankar 2022 [19]	America	Retrospective study	I–IV	79	38	12	29

Study	Age (years)	Myometrial invasion		Grade			Median follow-up (months)	Outcomes			
		<50%	≥50%	1	2	3		Prognostic indicator	Substantial LVSI	Local LVSI	No LVSI
Barnes 2021 [12]	66	117	208	157	101	67	35	4-year DFS	73%	89%	94%
								4-year OS	94%	91%	96%
Bosse 2015 [6]	67.8	278	648	673	137	116	160 for PORTEC-1 and 89 for PORTEC-2	5-years pelvic recurrence	15.3%	2.5%	1.7%
Matsuo 2015 [13]	NA	579	124	439	160	112	31.2	5-year distant recurrence	39.7%	10.1%	5.2%
								5-year pelvic recurrence	22.2%	2.0%	3.6%
Peters 2022 [14]	69	NA	NA	NA	NA	NA	102.6	5-year recurrence (stage I–II)	44.3%	23.0%	21.6%
								5-year recurrence (stage III–IV)	65.1%	33.3%	40.8%
Pifer 2020 [15]	63.6	310	128	287	112	27	NA	NA	NA	NA	NA
Pifer 2023 [16]	63.6	226	109	211	100	24	25.8	2-year locoregional DFS	91.4%	98.0%	92.5%
								2-year distant metastasis DFS	93.8%	93.3%	95.5%
								2-year OS	98.2%	98.1%	97.6%
Tortorella 2021 [17]	60	524	0	192	332	0	38	5-year OS	70.6%	100%	99.5%
								5-year DFS	56.5%	92.9%	93.6%
Restaino2020 [18]	63	712	542	898		360	36	5-year OS	68.3%	88.3%	88.9%
								5-year DFS	51.4%	77.6%	82.3%
Jaishankar 2022 [19]	68	51	28	NA	NA	NA	22.2	2-year locoregional DFS	71.4%	91.5%	
								2-year distant metastasis DFS	63.8%	90.2%	
								2-year OS	72.3%	95.4%	

Abbreviations: LVSI, lymph-vascular space invasion; DFS, disease-free survival; OS, overall survival.

2.35–4.793, $p = 0.000$, $I^2 = 10.7\%$) (Fig. 4) [6,12–14,18]. Furthermore, a comparison of focal and substantial LVSI groups found that substantial LVSI was more prone to pelvic recurrence (HR 1.82, 95% CI 1.05–3.15, $p = 0.033$, $I^2 = 0.0\%$) and distant recurrence (HR 2.21, 95% CI 1.48–3.28, $p = 0.000$, $I^2 = 0.0\%$) (Additional Fig. 1) [12,17,18].

3.5. The relationship between three-tiered scoring system for LVSI and lymph node involvement

The meta-analysis indicated that, compared to the no LVSI group, both focal and substantial LVSI groups had higher lymph involvement rates based on OR of 3.52 (95% CI 1.33–9.34, $p = 0.012$, $I^2 = 66.4\%$) for focal LVSI and 5.42 (95% CI 2.78–10.58, $p = 0.000$, $I^2 = 47.6\%$) for substantial LVSI (Additional Fig. 2) [13,15,16,18]. Pifer et al. observed that lymph nodes were involved in 3.3% of cases without LVSI, 7.5% of focal LVSI cases, and 15.2% of substantial LVSI cases [15]. Similarly, Matsuo et al. reported respective involvement rates of 4.1%, 36.4%, and 72.7% [13], while Restaino et al. found rates of 5%, 15%, and 33%, respectively [18]. Jaishankar et al. documented 0% involvement in the no LVSI group, 20% in the focal LVSI group, and 60% in the substantial LVSI group [19].

3.6. Subgroup analysis

3.6.1. Tumor stage

Most of the included studies in this review focused on early EC [6,12, 15–17]. In a study by Peter et al., the relationship between LVSI and prognosis was investigated in different tumor stage subgroups [14]. The results showed that substantial LVSI, compared to no LVSI, was associated with reduced OS in both stage I/II (HR 3.1, 95% CI 1.6–5.9, $p < 0.001$) and stage III/IV (HR 3.1, 95% CI 1.6–5.9, $p = 0.020$) patients.

Similarly, substantial LVSI was also associated with reduced 5-year cancer-specific survival (CSS) in stage III (HR 2.8, 95% CI 1.2–6.8, $p = 0.020$) and stage III/IV (HR 3.2, 95% CI 1.6–7.0, $p = 0.002$) patients, as well as decreased recurrence-free survival (RFS) in stage I/II (HR 2.8, 95% CI 1.3–6.2, $p = 0.011$) and stage III/IV (HR 2.9, 95% CI 1.4–6.0, $p = 0.003$) patients. On the other hand, focal LVSI was not found to be a significant risk factor for OS, CSS, or RFS [14].

3.6.2. Adjuvant radiotherapy

Several included studies investigated the therapeutic effects of adjuvant radiotherapy in patients with substantial LVSI. Bosse et al. observed that patients with substantial LVSI seemed to benefit from external beam radiotherapy (EBRT). The 5-year risk of pelvic regional recurrence in these women after EBRT was 4.3%, compared to 27.1% with vaginal brachytherapy and 30.7% with no additional treatment [6]. Restaino et al. found that adjuvant radiotherapy can improve DFS and OS in patients with substantial LVSI (median OS 93 versus 56 months, and median DFS 97 versus 19 months, for radiation therapy and no adjuvant, respectively, $p < 0.001$), regardless of histological grade, tissue type, lymph node metastasis (HR = 0.26, $p < 0.001$) [18]. Pifer et al. found that the 2-year locoregional-DFS was 93.0% in patients undergoing EBRT versus 87.8% in patients receiving vaginal brachytherapy or no radiation ($p = 0.473$). The 2-year distant metastasis-DFS were 93.5% and 94.1% ($p = 0.990$), while the 2-year OS rates were 98.4% and 100% ($p = 0.522$), respectively [16]. Barnes et al. reviewed outcomes of patients with early-stage endometrioid EC treated with adjuvant vaginal brachytherapy alone and found that substantial LVSI was the only risk factor predictive of pelvic (HR 7.2, 95%CI 1.0–51.6, $p = 0.048$) and distant failure (HR 4.4, 95%CI 1.2–16.3, $p = 0.048$) [1,2]. However, Barnes et al. discovered that even with the use of radiotherapy, substantial LVSI was associated with a higher risk of pelvic and distant failure [12].

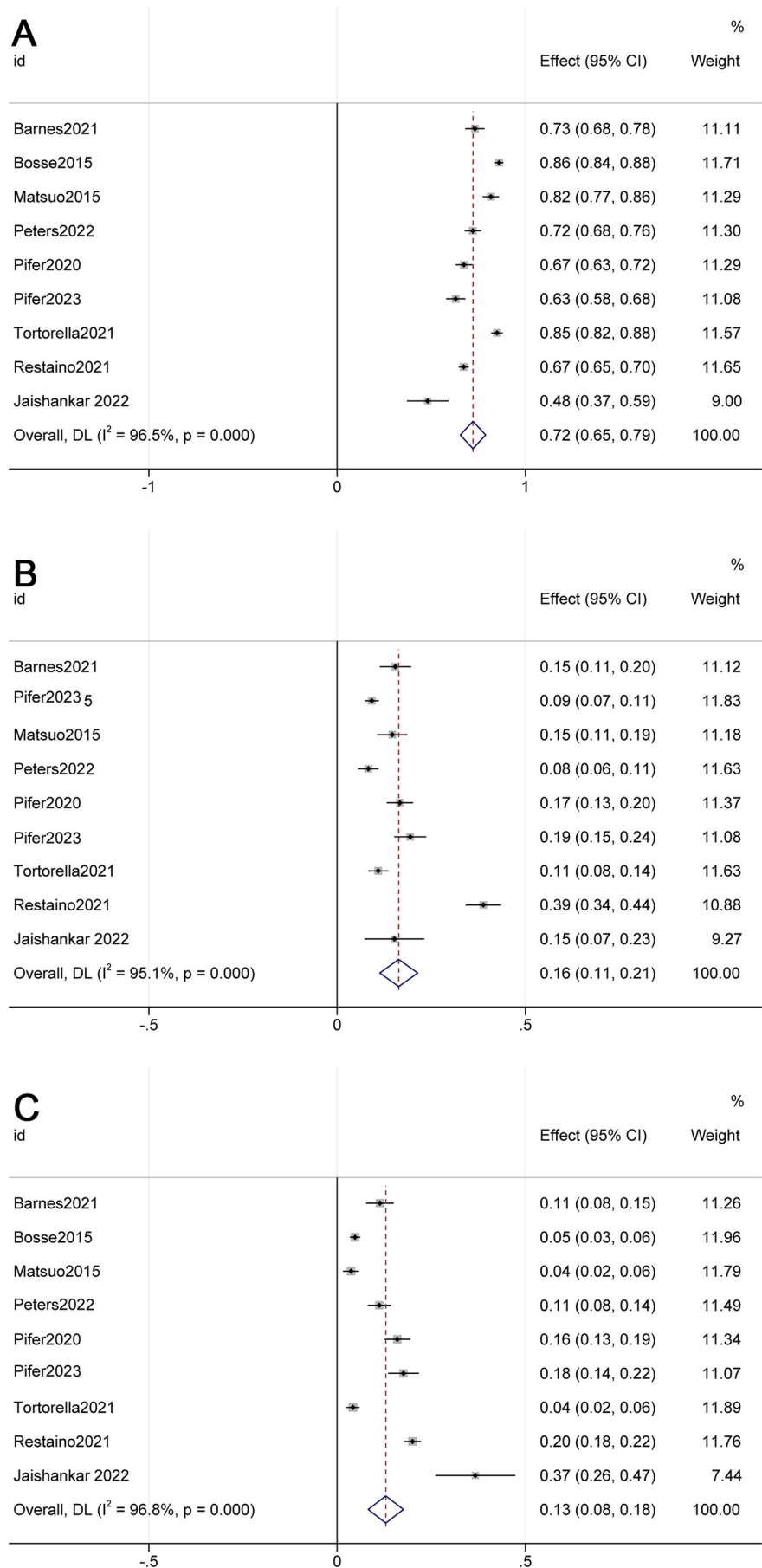


Fig. 2. Forest plot of pooled lymphovascular space invasion (LVSI) rates according to the three-tiered scoring system: (A) no LVSI, (B) focal LVSI, (C) substantial LVSI.

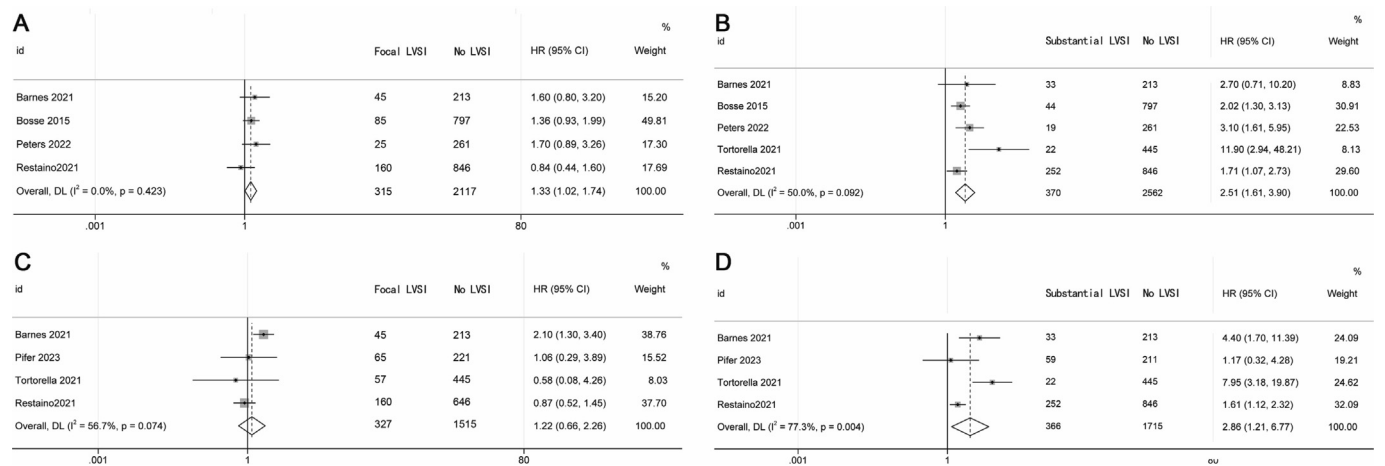


Fig. 3. Forest plot of survival outcomes according to the three-tiered scoring system for lymphovascular space invasion (LVSI) (compared to no LVSI): (A) overall survival for focal LVSI; (B) overall survival for substantial LVSI; (C) disease-free survival for focal LVSI; (D) disease-free survival for substantial LVSI.

3.6.3. Adjuvant chemotherapy

In 2015, Matsuo conducted a study analyzing the correlation between LVSI and prognosis in EC patients underwent chemotherapy or did not undergo chemotherapy. The results of the study indicated that in patients who did not receive chemotherapy, distant recurrence was found to be significantly associated with the extent of LVSI ($p < 0.001$). Further analysis revealed that extensive LVSI presented the highest rates of distant recurrence in comparison to those with no LVSI (HR 22.1, $p = 0.004$). However, in patients who underwent chemotherapy, there was no significant difference in the rates of distant recurrence among the LVSI groups analyzed ($p = 0.38$) [13].

4. Discussion

The presence of LVSI can be classified as no, focal, or substantial in a three-tier scoring system [6,20]. Focal LVSI was defined as a single focus of LVSI around the tumor, while substantial LVSI was defined as diffuse or multi-focal LVSI around the tumor [6]. Currently, there is no uniform definition for the number of vessels involved in substantial LVSI. FIGO 2023 defines substantial LVSI as ≥ 5 involved vessels, NCCN 2023 defines it as ≥ 4 involved vessels, and the American College of Pathologists (CAP) guidelines recommend it as ≥ 3 involved vessels [1,10,21]. Therefore, the definition of substantial LVSI varied across the studies included in our analysis.

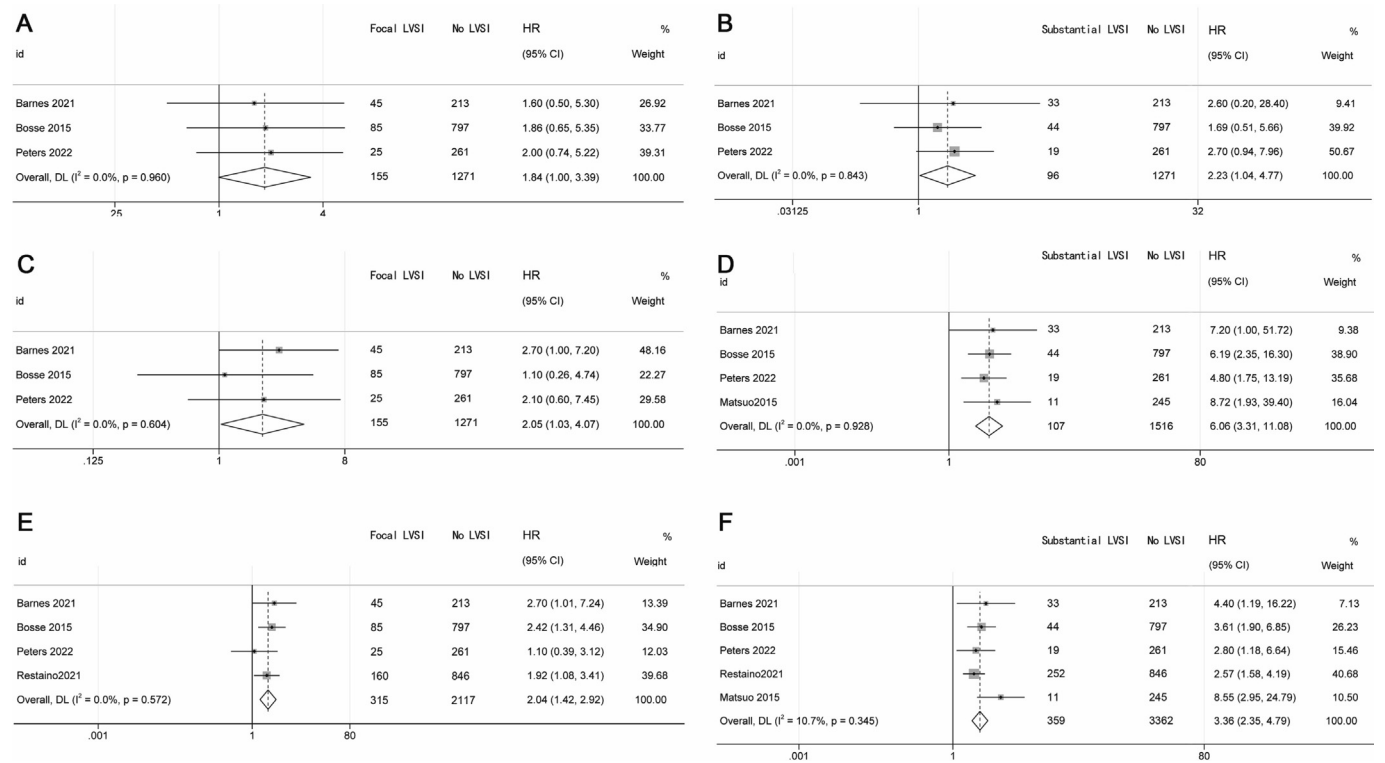


Fig. 4. Forest plot of recurrence outcomes according to the three-tiered scoring system for lymphovascular space invasion (LVSI) (compared to no LVSI): (A) local recurrence for focal LVSI; (B) local recurrence for substantial LVSI; (C) pelvic recurrence for focal LVSI; (D) pelvic recurrence for substantial LVSI; (E) distant recurrence for focal LVSI; (F) distant recurrence for substantial LVSI.

In this meta-analysis, the pooled incidence rates were 75% for no LVSI, 16% for focal LVSI, and 11% for substantial LVSI in EC patients. Studies have previously reported that the total incidence rate of LVSI ranges from 5.8% to 49.2% and the substantive LVSI ranges from 4.8% to 42% [6,12,15]. The incidence of LVSI is influenced by various factors. Firstly, the subjective judgments of observers and varying definitions of LVSI make it difficult to consistently reproduce the three-tier scoring system [6]. Secondly, patient conditions such as tumor stage, myometrial invasion, and pathological characteristics affect it [15]. Additionally, the incidence rate statistics can also be affected by differences in tumor specimen, including one slide per specimen or multiple slides per specimen [6,15]. Despite these challenges, the predictive value of the three-tier scoring system remains significant and has been incorporated into the new FIGO system. Consistent with previous studies [6,8,9,13,14,17] our findings suggested that LVSI is an independent prognostic factor for survival in EC patients. Comparing the focal and substantial LVSI groups to the no LVSI group, we observed poorer OS, worse DFS, and higher recurrence rates. Furthermore, the substantial LVSI group had a higher occurrence of pelvic and distant recurrence compared to the focal LVSI group.

Our study also revealed a higher incidence of lymph node involvement in the focal and substantial LVSI groups compared to the no LVSI group. This means that identifying the degree of LVSI may help determine the status of lymph node metastasis and avoid unnecessary lymph node dissection and related complications [13]. However, it remains unclear whether the worse prognosis is directly caused by the extent of LVSI itself or by the increased lymph node involvement [18]. The correlation between LVSI status and prognosis may vary depending on the presence or absence of lymph node involvement. For instance, Pifer et al. found no significant differences in DFS among subgroups with different LVSI statuses in lymph node-negative stage I EC [16]. Peters et al. reported a similar recurrence rate across different LVSI statuses in stage I/II lymph node-negative EC [14]. Barnes et al. showed that LVSI was not predictive of pelvic failure in patients who underwent lymph node assessment, but substantial LVSI was a prognostic factor in patients who did not undergo lymph node assessment [1,2]. Therefore, it may be important for patients who do not undergo lymph node dissection or evaluation to pay closer attention to their LVSI status.

Few articles analyzed the relationship between the three-tier scoring system and prognosis in different tumor stages or tumor histology. However, limited evidence suggested that regardless of tumor stage, substantial LVSI was a risk factor for prognosis. Substantial LVSI was a risk factor for local recurrence of stage I-II endometrioid EC, as well as for pelvic/paracortical lymph node recurrence, distant lymph node recurrence, distant metastasis, and OS reduction in non-stage I-II endometrioid cancer [14]. Compared with endometrioid cancer, LVSI was more common in type II EC, and substantial LVSI indicated the presence of lymph node metastasis and poorer clinical outcomes [19].

Several studies included in this review investigated the therapeutic effects of adjuvant radiotherapy in patients with substantial LVSI. The findings suggested that adjuvant radiotherapy seemed to improve survival and reduce recurrence in patients with substantial LVSI. Bosse et al. suggested that patients with stage I EC and substantial LVSI, especially grade 3 tumors, should receive adjuvant EBRT [6]. Both Bosse et al. and Barnes et al. recommended adjuvant pelvic EBRT for early-stage EC with substantial LVSI, while those with focal LVSI were advised to undergo adjuvant vaginal brachytherapy to reduce pelvic recurrence and minimize radiotherapy-related side effects [6,12]. This was in accordance with the ESMO guidelines, vaginal brachytherapy was recommended for intermediate risk EC, and EBRT was advised for high-intermediate risk EC [22].

Matsuo et al. observed that while there were differences in recurrence rates among different LVSI groups not receiving chemotherapy, there was no significant difference in distant recurrence rates among LVSI groups that did receive chemotherapy [13]. This may be due to

the association between LVSI and distant recurrence, and adjuvant chemotherapy potentially reducing recurrence stemming from small metastases to distant organs. The PORTEC-3 trial found that adjuvant chemotherapy in high-risk EC patients can improve survival [23]. However, although LVSI is one of the high-risk factors for EC, there is currently no research that analyzes whether adding chemotherapy is beneficial for improving the survival outcome of patients with LVSI [6].

A recent study has compared the prognoses of high-risk EC patients (LVSI being one of the high-risk factors) with different molecular classifications after adjuvant treatment. The study found that regardless of histologic type, molecular classification had a strong prognostic value for high-risk EC. Additionally, adjuvant treatment, including chemotherapy, can improve the prognosis of patients with p53 abnormalities. However, this study did not separately discuss the relationship between molecular classification and the prognosis of patients with LVSI [24]. An ongoing study (PORTEC-4a) is attempting to explore the combination of molecular classifications to guide the selection of adjuvant therapy for high-intermediate risk EC, but no conclusions have been reached yet [25]. Another recently published study reported that the presence of LVSI was associated with the recurrence of EC patients with a mismatch repair-deficient, p53 abnormal, and non-specific molecular profile. Additionally, in patients with a non-specific molecular profile, substantial LVSI was an independent predictor of recurrence [26]. Further research on molecular classification and current clinical pathological risk factors (low, low-intermediate, high-intermediate, high risk) will be necessary to guide adjuvant therapy in the future.

In this study, we followed a review protocol for study selection, data extraction, and analysis. Two independent reviewers performed study selection, data extraction, and assessed the risk of bias using standardized data extraction forms. Nonetheless, the study's findings should be interpreted in consideration of several limitations. First, most included studies were retrospective, which has inherent limitations. Second, the lack of a unified definition for substantial LVSI could cause misclassification bias, potentially affecting our conclusions. Third, subgroup analysis was limited due to insufficient studies. Lastly, we utilized a random-effects model for all analyses, which could result in down-weighting of large studies when statistical heterogeneity is present and assigning similar weighting to combined studies.

To conclude, the groups with focal and substantial LVSI exhibited worse survival rates and recurrence when compared to the group without LVSI. Specifically, the substantial LVSI group displayed an even greater likelihood of recurrence when compared to the focal LVSI group. The utilization of adjuvant radiotherapy and chemotherapy may prove beneficial for those in the substantial LVSI group. In the future, large, randomized studies and multicenter cohort studies will be necessary to further examine the impact of LVSI status on the prognosis of EC utilizing a three-tiered system for LVSI scoring to guide treatment decisions.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygyno.2024.01.046>.

Author contribution

Conceptualization, LH and HC; Methodology, LH and HC; Data Curation, HC and YC; Writing—Original Draft Preparation, LH and HC; Writing—Review and Editing, HC, XT and AZ. All authors have read and agreed to the published version of the manuscript.

CRediT authorship contribution statement

Ling Han: Writing – original draft, Methodology, Conceptualization. **Yali Chen:** Data curation. **Ai Zheng:** Writing – review & editing. **Xin Tan:** Writing – review & editing. **Hengxi Chen:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The author declares no conflict of interest.

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References

- [1] N.A. Rustum, C. Yashar, R. Arend, E. Barber, K. Bradley, R. Brooks, S.M. Campos, J. Chino, H.S. Chon, C. Chu, M.A. Crispens, S. Damast, C.M. Fisher, P. Frederick, D.K. Gaffney, R. Giuntoli, E. Han, J. Holmes, B.E. Howitt, J. Lea, A. Mariani, D. Mutch, C. Nagel, L. Nekhlyudov, M. Podoll, R. Salani, J. Schorge, J. Siedel, R. Sisodia, P. Soliman, S. Ueda, R. Urban, S.L. Wethington, E. Wyse, K. Zanotti, N.R. McMillian, S. Aggarwal, Uterine Neoplasms, Version 1.2023, NCCN Clinical Practice Guidelines in Oncology, J. Natl. Compr. Canc. Netw. 21 (2023) 181–209, <https://doi.org/10.6004/jcnccn.2023.0006>.
- [2] R.L. Siegel, K.D. Miller, N.S. Wagle, A. Jemal, Cancer statistics, CA Cancer, J. Clin. 73 (2023) 17–48, <https://doi.org/10.3322/caac.21763>.
- [3] R.L. Siegel, K.D. Miller, N.S. Wagle, A. Jemal, Cancer statistics, CA Cancer, J. Clin. 67 (2017) 7–30, <https://doi.org/10.3322/caac.21387>.
- [4] N. Singh, L. Hirschowitz, R. Zaino, I. Alvarado-Cabrero, M.A. Duggan, R. Ali-Fehmi, E. Euscher, J.L. Hecht, L.C. Horn, O. Ioffe, X. Matias-Guiu, W.G. McCluggage, Y. Mikami, J. Ordi, V. Parkash, M.R. Qudus, C.M. Quick, A. Scablon, C. Zaloudek, M. Nucci, A. Malpica, E. Oliva, Pathologic prognostic factors in endometrial carcinoma (other than tumor type and grade), Int. J. Gynecol. Pathol. 38 (2019) S93–S113, doi: 10.1097/PGP.0000000000000524.
- [5] N. Colombo, C. Creutzberg, F. Amant, T. Bosse, A. González-Martín, J. Ledermann, C. Marth, R. Nout, D. Querleu, M.R. Mirza, C. Sessa, Cancer: diagnosis, treatment and follow-up, Ann. Oncol. 27 (2016) 16–41, <https://doi.org/10.1093/annonc/mdv484>.
- [6] T. Bosse, E.E. Peters, C.L. Creutzberg, I.M. Jürgenliemk-Schulz, J.J. Jobsen, J.W. Mens, L.C. Lutgens, E.M.V.D. Steen-Banasik, V.T. Smit, R.A. Nou, Substantial lymphovascular space invasion (LVSI) is a significant risk factor for recurrence in endometrial cancer – A pooled analysis of PORTEC 1 and 2 trials, Eur. J. Cancer 51 (2015) 1742–1750, <https://doi.org/10.1016/j.ejca.2015.05.015>.
- [7] K. Ståhlberg, M. Bjurberg, C. Borgfeldt, J. Carlson, P. Dahm-Kähler, P. A. Flöter-Rådestad, K. Hellman, E. Hjerpe, E. Holmberg, P. Kjølhed, J. Marcickiewicz, P. Rosenberg, B. Tholander, E. Åvall-Lundqvist, T. Höglberg, Lymphovascular space invasion as a predictive factor for lymph node metastases and survival in endometrioid endometrial cancer – a Swedish Gynecologic Cancer Group (SweGCG) study, Acta. Oncol. 58 (2019) 1628–1633, doi: 10.1080/0284186X.2019.1643036.
- [8] D.J. O'Brien, G. Flannelly, E.E. Mooney, M. Foley, Lymphovascular space involvement in early stage well-differentiated endometrial cancer is associated with increased mortality, Br. J. Obstet. Gynaecol. 116 (2009) 991–994, doi:10.1111/j.1471-0528.2009.02162.x. [9] T. Hachisuga, T. Kaku, K. Fukuda, F. Eguchi, M. Emoto, T. Kamura, T. Iwasaka, T. Kawarabayashi, H. Sugimori, M. Mori, The grading of lymphovascular space invasion in endometrial carcinoma, Cancer 86 (1999) 2090–2097.
- [9] T. Hachisuga, T. Kaku, K. Fukuda, F. Eguchi, M. Emoto, T. Kamura, T. Iwasaka, T. Kawarabayashi, H. Sugimori, M. Mori, The grading of lymphovascular space invasion in endometrial carcinoma, Cancer 86 (1999) 2090–2097.
- [10] J.S. Berek, X. Matias-Guiu, C. Creutzberg, C. Fotopoulou, D. Gaffney, S. Kehoe, K. Lindemann, D. Mutch, N. Concin, Endometrial Cancer Staging Subcommittee, FIGO Women's Cancer Committee, FIGO staging of endometrial cancer: 2023, Int. J. Gynaecol. Obstet. 162 (2023) 383–394, doi:10.1002/ijgo.14923. A. León-Castillo, S.M. de Boer, M.E. Powell, L.R. Mileschkin, H.J. Mackay, A. Leary, H.W. Nijman, N. Singh, P.M. Pollock, P. Bessette, A. Fyles, C. Haie-Meder, V.T.H.B.M. Smit, R.J. Edmondson, H. Putter, H.C. Kitchener, E.J. Crosbie, M. de Bruyn, R.A. Nout, N. Horeweg, C.L. Creutzberg, T. Bosse, TransPORTEC consortium, Molecular classification of the PORTEC-3 trial for high-risk endometrial cancer: Impact on prognosis and benefit from adjuvant therapy, J. Clin. Oncol. 38 (2020) 3388–3397, doi: 10.1200/JCO.20.00549.
- [11] A. Liberati, D.G. Altman, J. Tetzlaff, C. Mulrow, P.C. Gøtzsche, J.P. Ioannidis, M. Clarke, P.J. Devereaux, J. Kleijnen, D. Moher, The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration, J. Clin. Epidemiol. 339 (2009), b2700.
- [12] E.A. Barnes, K. Martell, C. Parra-Herran, A.S. Taggar, E. Donovan, E. Leung, Substantial lymphovascular space invasion predicts worse outcomes in early-stage endometrioid endometrial cancer, Brachytherapy 20 (2021) 527–535, <https://doi.org/10.1016/j.brachy.2020.12.006>.
- [13] K. Matsuo, J. Garcia-Sayre, F. Medeiros, J.K. Casabar, H. Machida, A. Moeini, L.D. Roman, Impact of depth and extent of lymphovascular space invasion on lymph node metastasis and recurrence patterns in endometrial cancer, J. Surg. Oncol. 112 (2015) 669–676, <https://doi.org/10.1002/jso.24049>.
- [14] E.E.M. Peters, A. León-Castillo, E. Hogdall, M. Boenelycke, V.T.H.B.M. Smit, C. Hogdall, C.L. Creutzberg, T. Bosse, R.A. Nout, G. Ørtoft, Substantial lymphovascular space invasion is an adverse prognostic factor in high-risk endometrial cancer, Int. J. Gynecol. Pathol. 41 (2022) 227–234, <https://doi.org/10.1097/PGP.0000000000000805>.
- [15] P.M. Pifer, R. Bhargava, A.K. Patel, D.C. Ling, J.A. Vargo, B.C. Orr, P. Sukumvanich, M.B. Courtney-Brooks, M.M. Boisen, J.L. Berger, S. Taylor, A.B. Olawaiye, J.T. Comer, J.L. Lesnock, R.P. Edwards, S. Beriwal, Is the risk of substantial LVSI in stage I endometrial cancer similar to PORTEC in the north American population? – a single-institution study, Gynecol. Oncol. 159 (2020) 23–29, <https://doi.org/10.1016/j.ygyno.2020.07.024>.
- [16] P.M. Pifer, S. Jaishankar, R. Bhargava, M.D. Schad, A. Keller, H.B. Musunuru, M. Cohen, P. Sukumvanich, M. Courtney-Brooks, M. Boisen, J.L. Berger, A. Olawaiye, J. Lesnock, R. Edwards, S. Taylor, J.A. Vargo, S. Beriwal, Is substantial lymphovascular space invasion prognostic in patients with pathologically lymph node-negative endometrial cancer? Int. J. Radiat. Oncol. Biol. Phys. 117 (2023) 148–153, <https://doi.org/10.1016/j.ijrobp.2023.02.053>.
- [17] L. Tortorella, S. Restaino, G.F. Zannoni, G. Vizzielli, V. Chiantera, S. Cappuccino, A. Gioè, E. La Fera, G. Dinoi, G. Angelico, G. Scambia, F. Fanfani, Substantial lymphovascular space invasion (LVSI) as predictor of distant relapse and poor prognosis in low-risk early-stage endometrial cancer, J. Gynecol. Oncol. 32 (2021) e11, doi:10.3802/jgo.2021.32.e11.
- [18] S. Restaino, L. Tortorella, G. Dinoi, G.F. Zannoni, A. Baroni, I. Capasso, E. Distefano, G. Sozzi, V. Chiantera, G. Scambia, F. Fanfani, Semiquantitative evaluation of lymphovascular space invasion in patients affected by endometrial cancer: prognostic and clinical implications, Eur. J. Cancer 142 (2021) 29–37, <https://doi.org/10.1016/j.ejca.2020.10.011>.
- [19] S. Jaishankar, P.M. Pifer, R. Bhargava, A. Keller, H.B. Musunuru, A.K. Patel, P. Sukumvanich, M. Boisen, J.L. Berger, S. Taylor, M. Courtney-Brooks, A. Olawaiye, J. Lesnock, R. Edwards, J.A. Vargo, S. Beriwal, Is substantial lymphovascular space invasion prognostic for clinical outcomes in type II endometrial cancer? Clin. Oncol. 34 (2022) 452–458, <https://doi.org/10.1016/j.clon.2022.02.018>.
- [20] WHO Classification of Tumours Editorial Board, Female Genital Tumours, WHO Classification of Tumours, IARC Press, 2020.
- [21] G.K. Uma, A.C. Barbara, The CAP Cancer and CAP pathology electronic reporting committees, Protocol for the Examination of Specimens From Patients with Carcinoma and Carcinosarcoma of the Endometrium, <https://www.cap.org/protocols-and-guidelines/cancer-reporting-tools/cancer-protocol-templates>.
- [22] A. Oaknin, T.J. Bosse, C.L. Creutzberg, G. Giromelli, P. Harter, F. Joly, D. Lorusso, C. Marth, V. Makker, M.R. Mirza, J.A. Ledermann, N. Colombo, ESMO guidelines committee, endometrial cancer: ESMO clinical practice guideline for diagnosis, treatment and follow-up, Ann. Oncol. 33 (2022) 860–877, <https://doi.org/10.1016/j.annonc.2022.05.009>.
- [23] S.M. de Boer, M.E. Powell, L. Mileschkin, D. Katsaros, P. Bessette, C. Haie-Meder, P.B. Ottevanger, J.A. Ledermann, P. Khaw, R. D'Amico, A. Fyles, M.H. Baron, I.M. Jürgenliemk-Schulz, H.C. Kitchener, H.W. Nijman, G. Wilson, S. Brooks, S. Gribaudo, D. Provencher, C. Hanzen, R.F. Kruitwagen, V.T.H.B.M. Smit, N. Singh, V. Do, A. Lissoni, R.A. Nout, A. Feeney, K.W. Verhoeven-Adema, H. Putter, C.L. Creutzberg, PORTEC Study Group, Adjuvant chemoradiotherapy versus radiotherapy alone in women with high-risk endometrial cancer (PORTEC-3): patterns of recurrence and post-hoc survival analysis of a randomised phase 3 trial, Lancet, Oncol. 20 (2019) 1273–1285.
- [24] A. León-Castillo, S.M. de Boer, M.E. Powell, L.R. Mileschkin, H.J. Mackay, A. Leary, H.W. Nijman, N. Singh, P.M. Pollock, P. Bessette, A. Fyles, C. Haie-Meder, V.T.H.B.M. Smit, R.J. Edmondson, H. Putter, H.C. Kitchener, E.J. Crosbie, M. de Bruyn, R.A. Nout, N. Horeweg, C.L. Creutzberg, T. Bosse, TransPORTEC consortium, Molecular classification of the PORTEC-3 trial for high-risk endometrial cancer: Impact on prognosis and benefit from adjuvant therapy, J. Clin. Oncol. 38 (2020) 3388–3397, <https://doi.org/10.1200/JCO.20.00549>.
- [25] A.S.V.M. Van den Heerik, N. Horeweg, R.A. Nout, L.C.H.W. Lutgens, E.M. van der Steen-Banasik, G.H. Westerveld, H.A. van den Berg, A. Slot, F.L.A. Koppe, S. Kommos, J.W.M. Mens, M.E. Nowee, S. Bijmolt, D. Cibula, T.C. Stam, I.M. Jürgenliemk-Schulz, A. Snyers, M. Hamann, A.G. Zwanenburg, V.L.M.A. Coen, K. Vandecasteele, C. Gillham, C. Chagari, K.W. Verhoeven-Adema, H. Putter, W.B. van den Hout, B.G. Wortman, H.W. Nijman, T. Bosse, C.L. Creutzberg, PORTEC-4a: International randomized trial of molecular profile-based adjuvant treatment for women with high-intermediate risk endometrial cancer, Int. J. Gynecol. Cancer 30 (2020 2002–2007), <https://doi.org/10.1136/ijgc-2020-001929>.
- [26] F. Siegenthaler, E. Epstein, C.A. Büchi, A. Gmür, F.A.C.M. Saner, T.T. Rau, J.W. Carlson, M.D. Mueller, S. Imboden, Prognostic value of lymphovascular space invasion according to the molecular subgroups in endometrial cancer, Int. J. Gynecol. Cancer 33 (2023) 1702–1707, <https://doi.org/10.1136/ijgc-2023-004606>.